THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — SCIENCE FULL SYLLABUS PAPER SET 04

Full Syllabus (Chapters 1–13) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The symbol (aq) in a chemical equation means: 1

(a) aqueous solution **(b)** acid **(c)** air **(d)** aquatic

2. Antacids such as milk of magnesia contain: 1

(a) $MgSO_4$ **(b)** $MgCO_3$ only **(c)** $MgCl_2$ **(d)** $Mg(OH)_2$

3. A student states: “The pH of rain water is always exactly 7.” The correct response is: 1

(a) Correct

(b) Incorrect — when it is acid rain its pH is less than 5.6

(c) It is always 9

(d) Cannot be determined

4. The allotrope of carbon that is the hardest natural substance is: 1

(a) diamond **(b)** fullerene **(c)** coal **(d)** graphite

5. Magnesium reacts with hot water to produce: 1

(a) magnesium oxide and oxygen **(b)** only hydrogen

(c) magnesium hydroxide and hydrogen **(d)** magnesium chloride

6. Cyclohexane has the formula: 1

(a) C_6H_{12} **(b)** C_6H_{14} **(c)** C_6H_6 **(d)** C_6H_{10}

7. Substitution of hydrogen in methane by chlorine requires: 1

(a) conc. sulphuric acid **(b)** nickel catalyst

(c) sunlight **(d)** darkness

8. A student states: “Veins have valves to prevent backflow of blood.” The correct response is: 1

(a) Correct **(b)** Incorrect

(c) Only arteries have valves **(d)** Cannot be determined

9. The longest fibre of a nerve cell is the: 1

(a) cell body **(b)** synapse **(c)** dendrite **(d)** axon

10. Menstruation lasts about: 1

(a) 2 to 8 days **(b)** one hour **(c)** 28 days **(d)** 15 days

11. The fluid that carries sperms is formed by secretions from: 1

(a) seminal vesicles and prostate **(b)** ovaries

(c) kidneys **(d)** liver

12. The ratio of tall to short plants in the F2 of a monohybrid cross is: 1

(a) 1 : 1 **(b)** 1 : 2 : 1 **(c)** 3 : 1 **(d)** 9 : 3 : 3 : 1

13. Which genotype will give short plants? 1

(a) Tt **(b)** tt **(c)** TT **(d)** all of these

14. The lens formula is: 1

(a) $\frac{1}{v}+\frac{1}{u}=\frac{1}{f}$ **(b)** $v-u=f$ **(c)** $\frac{1}{v}-\frac{1}{u}=\frac{1}{f}$ **(d)** $uv=f$

15. The colour that is scattered the most by air molecules is: 1

(a) blue/violet **(b)** orange **(c)** yellow **(d)** red

16. If the current through a resistor is doubled, the heat produced becomes: 1

(a) same **(b)** four times **(c)** double **(d)** half

17. The resistivity of a material depends on: 1

(a) area **(b)** length

(c) nature of material and temperature **(d)** shape

18. Short circuit causes: 1

(a) zero current **(b)** a sudden large current

(c) no effect **(d)** low voltage

19. The international agreement in 1987 under UNEP aimed to: 1

(a) ban plastic **(b)** increase ozone

(c) reduce CO_2 **(d)** freeze CFC production at 1986 levels

20. Which organism occupies the highest trophic level in the chain grass → grasshopper → frog → snake → eagle? 1

(a) frog **(b)** grass **(c)** snake **(d)** eagle

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 10–15 words each.

21. Balance: 2

(i) $H_2 + O_2 \longrightarrow H_2O$

(ii) $Mg + HCl \longrightarrow MgCl_2 + H_2$

22. Why does a solution of NaCl not change the colour of litmus? 2

23. Name the end products of digestion of carbohydrates, proteins and fats (any two). 2

24. What are effectors? Give one example. 2

25. Name two plants propagated by grafting or layering. 2

26. Why do offspring resemble their parents but are not identical? 2

27. The power of a lens is +4 D. Find its focal length and type. 2

28. A student says that the earth wire carries current all the time. Justify or refute. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. What is a redox reaction? Identify the oxidising and reducing agents in — 3
(i) $\mathrm{Fe_2O_3 + 2\,Al \longrightarrow Al_2O_3 + 2\,Fe}$
(ii) $\mathrm{2\,H_2S + SO_2 \longrightarrow 3\,S + 2\,H_2O}$.

30. An element X (atomic number 12) reacts with element Y (atomic number 17). — 3
(i) What type of bond is formed?
(ii) Write the formula.
(iii) Is the compound soluble in water?

31. Distinguish between phototropism and geotropism with examples. — 3

32. What are the agents of pollination? Give one example of each. — 3

33. Why does the Sun appear red at sunrise and sunset but white at noon? — 3

34. What happens to the force on a conductor when — 3
(i) current is increased
(ii) field is reversed
(iii) the conductor is made parallel to field?

35. What happens to the concentration of $\mathrm{H_3O^+}$ ions when excess base is added to an acid solution? What happens to the pH? Why? — 3

OR

State three uses of common salt (sodium chloride). — 3

36. Explain how energy is transferred in a food chain of grass → rabbit → fox, using the 10% law (take 10 000 J at grass). — 3

OR

Distinguish between a food chain and a food web (three points). — 3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) Describe image formation by a convex lens when the object is — 5
(a) between F and 2F
(b) between F and O.
(ii) A 3 cm object is placed 20 cm from a convex lens of focal length 10 cm. Find the image size.

OR

(i) What is meant by power of a lens? — 5
(ii) Find the power of a combination of a convex lens of focal length 25 cm and a concave lens of focal length 10 cm.
(iii) Is the combination converging?

38. (i) Explain covalent bonding with $\mathrm{H_2}$, $\mathrm{O_2}$ and $\mathrm{N_2}$ examples. — 5
(ii) Why do covalent compounds have low melting points?

OR

(i) Describe the chemical properties of ethanol (reaction with Na, dehydration). — 5
(ii) Harmful effects of alcohol.
(iii) What is the use of ethanol as a fuel?

MASTERMIND

39. (i) Why do the filaments of bulbs glow? 5
(ii) Why is an inert gas filled in bulbs?
(iii) Find the heat developed in 5 minutes by a bulb of 220 V, 60 W.

OR

Case study: A student measured the resistance of wires A and B of the same material. A is 1 m long, 1 mm² area; B is 2 m long, 0.5 mm² area. 5
(i) Which has more resistance?
(ii) Find the ratio R_B/R_A.
(iii) What is resistivity?
(iv) Why is resistivity the same for both?
(v) Name a material with very low resistivity.

40. **Case study:** A student kept a beaker of yeast in sugar solution in a warm place; bubbles appeared and the solution smelled of alcohol. 5
(i) Name the process.
(ii) Write the products.
(iii) Where in the cell does it take place?
(iv) How is it different from respiration in our muscles during exercise?
(v) Which process gives more energy?

OR

Case study: Activity: freshly prepared lime water in two test tubes; air is pumped into one and exhaled air is blown into the other. 5
(i) In which tube does lime water turn milky faster?
(ii) Why?
(iii) What is the gas responsible?
(iv) What is the composition difference between inhaled and exhaled air?
(v) Name the process in which energy is released from food.

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND